The documentation and process conversion measures necessary to comply with this revision shall be completed by 18 Feb 94.

INCH-POUND

MIL-S-19500/456C
18 November 1993
SUPERSEDING
MIL-S-19500/456B
17 September 1979

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER
TYPE 2N5302 AND 2N5303, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, high-power transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

	P_T 1/		V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$	
	T_A = +25°C	T_C = +100°C								
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W Min	°C/W Max
2N5302	5	115	60	60	5	7.5	30	-65 to +200		.875
2N5303	5	115	80	80	5	7.5	20	-65 to +200		.875

1/ Derate linearly 1.14 W/°C above T_C = 100°C.
Derate linearly 28.57 mW/°C above T_A = 25°C.

1.4 Primary electrical characteristics.

	h_{FE2} 1/	h_{FE2} 1/	$\|h_{FE}\|$	$V_{BE(sat)2}$ 1/		$V_{CE(sat)2}$ 1/		C_{obo}	Pulse response	
	V_{CE} = 2 V dc, I_C = 15 A dc	V_{CE} = 2 V dc, I_C = 10 A dc	V_{CE} = 10 V dc, I_C = 1 A dc, f = 1 MHz	I_C = 15 A dc, I_B = 1.5 A dc		I_C = 15 A dc, I_B = 1.5 A dc		V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	t_{on}	t_{off}
	2N5302	2N5303	V dc	2N5302 V dc	2N5303 V dc	2N5302 V dc	2N5303 V dc	pF	µs	µs
MIN	15	15	2							
MAX	60	60	40	1.8	2	1	1.5	800	1.1	3.0

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington, Pike, Dayton, OH 45444-5270 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 11, T-4A.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.3 Screening. Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CEX1} and h_{FE2}	Not applicable
11	I_{CEX1} and h_{FE2}; $\Delta I_{CEX1} \leq$ 100 percent of initial value or 50 µA dc, whichever is greater. $\Delta h_{FE2} \leq \pm$20 percent of initial value.	I_{CEX1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; $\Delta I_{CEX1} \leq$ 100 percent of initial value or 50 µA dc whichever is greater; $\Delta h_{FE2} \leq \pm$15 percent of initial value.	Subgroup 2 of table I herein; $\Delta I_{CEX1} \leq$ 100 percent of initial value or 100 µA dc, whichever is greater; $\Delta h_{FE2} \leq$ 20 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5°C ±12.5°C; V_{CE} = 25 V dc ±5 V dc; $T_A \leq$ 100°C

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table II herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
4	1037	V_{CB} = 20 V dc; P_D = 5 W at T_A = room ambient as defined in 4.5 of MIL-STD-750; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling on devices shall be permitted.
5	1027	V_{CB} = 20 V dc; T_A = +125°C ±25°C for 96 hours; P_T = 5 W at T_A = +125°C or adjusted as required by the chosen T_A to give an average lot, T_J = +275°C.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
3	1037	$V_{CB} \geq$ 10 V dc; ΔT_J between cycles ≥ 100°C; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling on devices shall be permitted.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) shall be in accordance with table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
6	1037	$V_{CB} \geq 10$ V dc; ΔT_J between cycles $\geq 100°C$; $t_{on} = t_{off} = 3$ minutes minimum for 6,000 cycles. No heat sink or forced air cooling on devices shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 3.14 A dc.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be $25°C \leq T_R \leq 75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 0.875°C/W.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified.

c. Type designation and product assurance level.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector - emitter breakdown voltage	3011	Bias condition D; I_C = 200 mA dc; pulsed (see. 4.5.1)	$V_{(BR)CEO}$			
2N5302				60		V dc
2N5303				80		V dc
Collector - emitter cutoff current	3041	Bias condition D;	I_{CEO}			
2N5302		V_{CE} = 60 V dc			10.0	µA dc
2N5303		V_{CE} = 80 V dc			5.0	µA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		5.0	µA dc
Collector - emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc;	I_{CEX1}			
2N5302		V_{CE} = 60 V dc			5.0	µA dc
2N5303		V_{CE} = 80 V dc			5.0	µA dc
Collector - base cutoff current	3036	Bias condition D;	I_{CBO}			
2N5302		V_{CE} = 60 V dc			1.7	V dc
2N5303		V_{CE} = 80 V dc			1.8	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 A dc; I_B = 1 A dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$			
Base - emitter saturated voltage	3066	Test condition A; I_C = 15 A dc; I_B = 1.5 A dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$			
2N5302					2.5	V dc
2N5303					2.5	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 20 A dc; pulsed (see 4.5.1);	$V_{BE(sat)3}$			
2N5302		I_B = 2 A dc			1.8	V dc
2N5303		I_B = 4 A dc			1.5	V dc
Base - emitter voltage (nonsaturated)	3066	Test condition B; V_{CE} = 2 V dc; pulsed (see 4.5.1);	V_{BE1}			
2N5302		I_C = 15 A dc				
2N5303		I_C = 10 A dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Base - emitter voltage (nonsaturated)	3066	Test condition B; V_{CE} = 4 V dc; pulsed (see 4.5.1);	V_{BE2}			
2N5302		I_C = 30 A dc			3.0	V dc
2N5303		I_C = 20 A dc			2.5	V dc
Collector - emitter saturated voltage	3071	I_C = 10 A dc; I_B = 1 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N5302					0.75	V dc
2N5303					1.0	V dc
Collector - emitter saturated voltage	3071	I_C = 15 A dc; I_B = 1.5 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N5302					1.0	V dc
2N5303					1.5	V dc
Collector - emitter saturated voltage	3071	I_C = 20 A dc; pulsed (see 4.5.1);	$V_{CE(sat)3}$			
2N5302		I_B = 2 A dc			2.0	V dc
2N5303		I_B = 4 A dc			2.0	V dc
Collector - emitter saturated voltage	3071	I_C = 30 A dc; pulsed (see 4.5.1);	$V_{CE(sat)4}$			
2N5302 (only)		I_B = 6 A dc			3.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE1}	40		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; pulsed (see 4.5.1);	h_{FE2}			
2N5302		I_C = 15 A dc		15	60	
2N5303		I_C = 10 A dc		15	60	
Forward-current transfer ratio	3076	V_{CE} = 4 V dc; pulsed (see 4.5.1);	h_{FE3}			
2N5302		I_C = 30 A dc		5		
2N5303		I_C = 20 A dc		5		
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc;	I_{CEX2}		50	µA dc
2N5302		V_{CE} = 60 V dc				
2N5303		V_{CE} = 80 V dc				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued.						
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; pulsed (see 4.5.1);	h_{FE4}			
2N5302		I_C = 15 A dc		7		
2N5303		I_C = 10 A dc		7		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 A dc; f = 1 kHz	h_{fe}	40	240	
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1 A dc; f = 1 MHz	$\|h_{fe}\|$	2	40	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz < f < = 1 MHz	C_{obo}		800	pF
Switching parameters:						
Pulse delay time		see figure 1	t_d		0.2	µs
Pulse rise time		see figure 1	t_r		0.9	µs
Pulse storage time		see figure 1	t_s		2.0	µs
Pulse fall time		see figure 1	t_f		1.0	µs
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = 25°C; power application time ≥ 1 sec, 1 cycle (see figure 2)				
Test 1						
2N5302		V_{CE} = 6.67 V dc; I_C = 30 A dc				
2N5303		V_{CE} = 10 V dc; I_C = 20 A dc				
Test 2						
2N5302, 2N5303		V_{CE} = 20 V dc; I_C = 10 A dc				

See footnote at end of table.

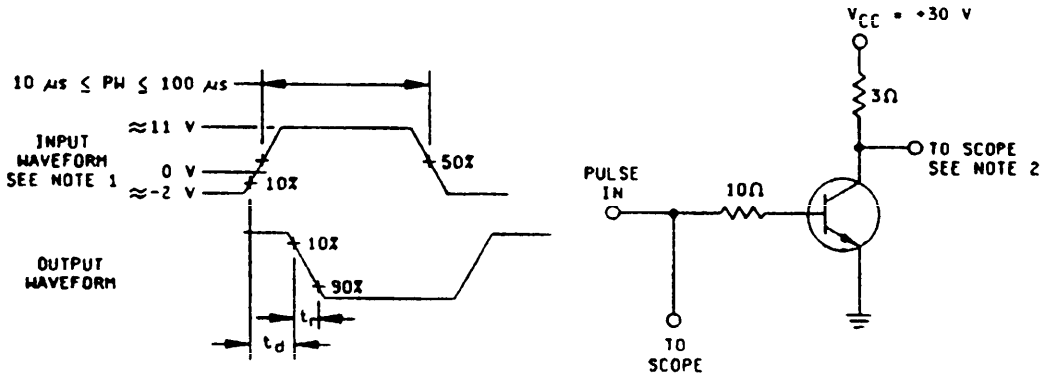
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TABLE I. Group A inspection - Continued.

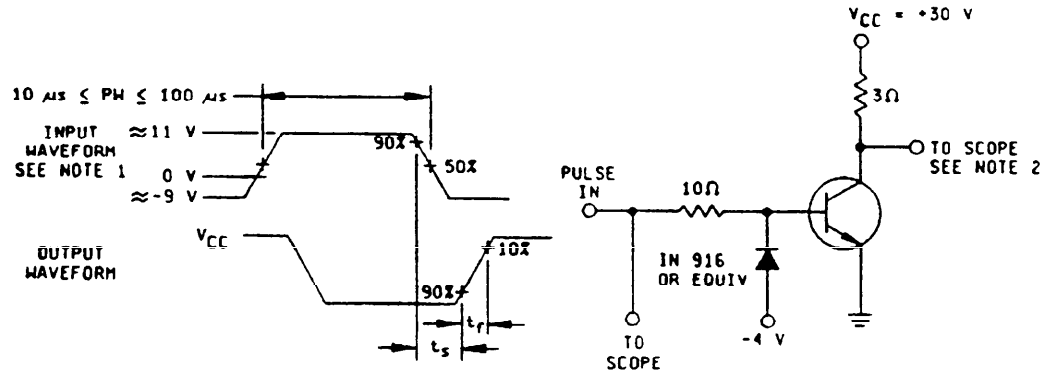
Inspection	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued.						
Test 3						
2N5302, 2N5303		V_{CE} = 40 V dc; I_C = 3 A dc				
Test 4						
2N5302		V_{CE} = 50 V dc; I_C = 600 mA dc				
2N5303		V_{CE} = 60 V dc; I_C = 600 mA dc				
Safe operating area (clamped switching)		T_A = 25°C; V_{CE} = 15 V dc; (see figures 3 and 4)				
2N5302		Clamp voltage = 60 V dc; I_C = 30 A dc				
2N5303		Clamp voltage = 80 V dc; I_C = 20 A dc				
Electrical Measurements		See table I, group A, subgroup 2				
Subgroups 6 and 7						
Not applicable						

For sampling plan see MIL-S 19500.

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t_d AND t_r TIME TEST CIRCUIT



t_s AND t_f TIME TEST CIRCUIT

NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{IN} ≥ 100 kΩ; C_{IN} ≤ 12 pF; rise time ≤ 0.2 ns.

FIGURE 1. Pulse response test circuit.

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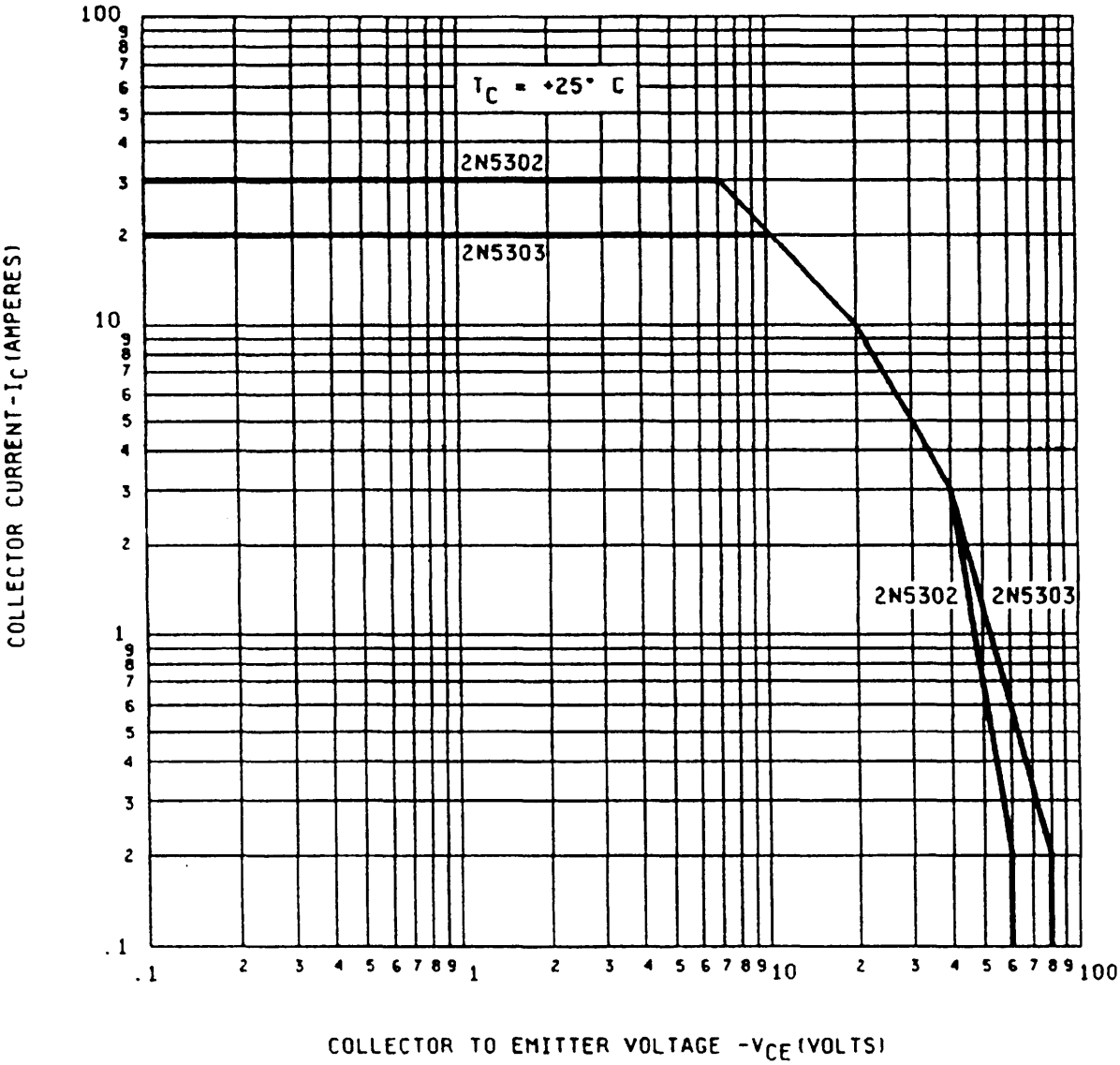


FIGURE 2. Maximum safe operating area graph dc.

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See footnote at end of table.

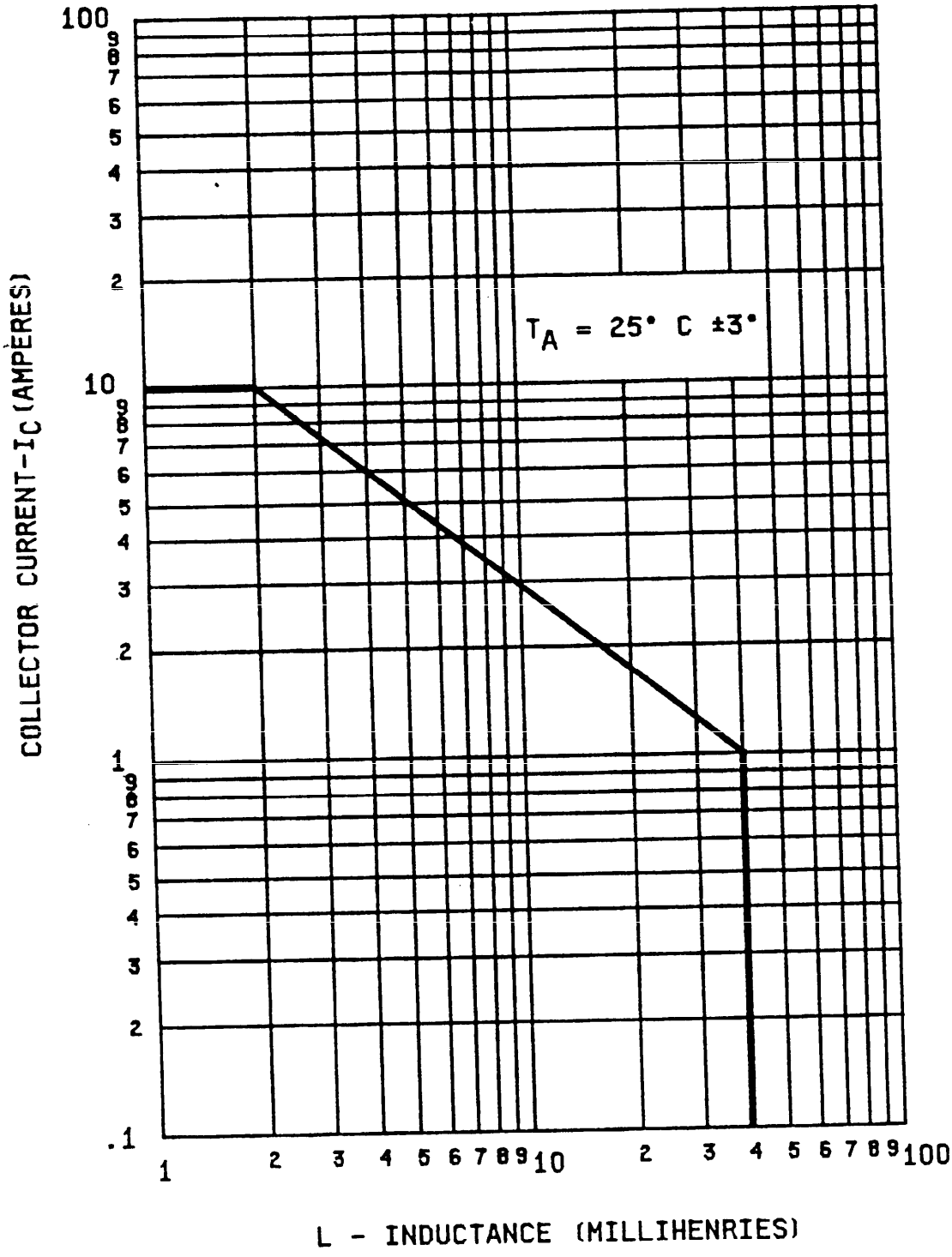
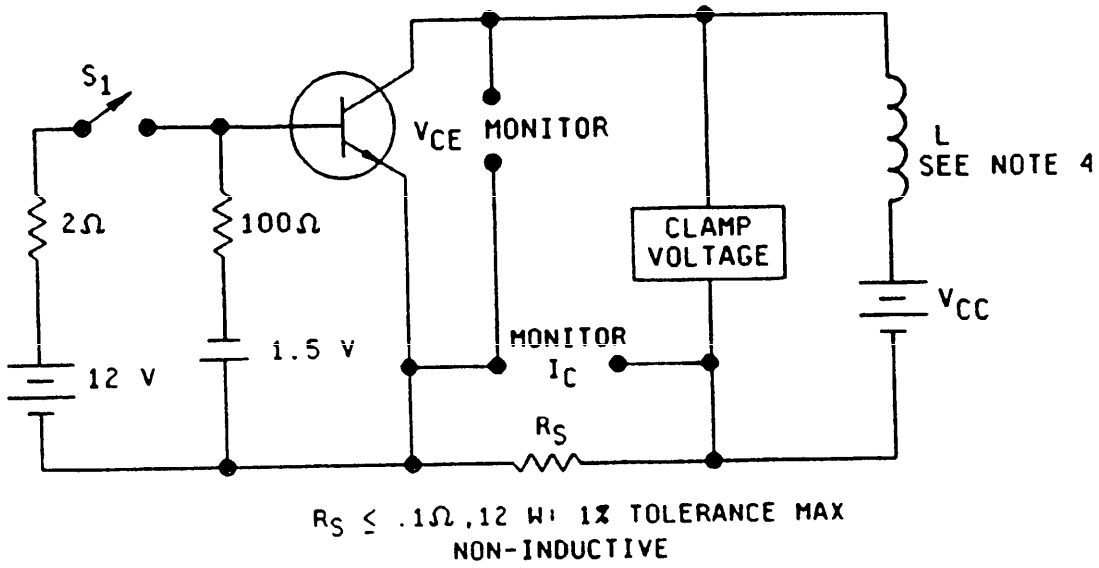


FIGURE 3. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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*L = 2.0 mH (2 each 1 mH, 50 A, .001 Ω, Sanford Miller CK-50, or equivalent).

Procedure:

1. With switch S1 closed, set the specified test conditions.
2. Open S1. Device fails if the clamp voltage if not reached.
3. Perform specified endpoints tests.

FIGURE 4. Clamped inductive sweep test circuit.

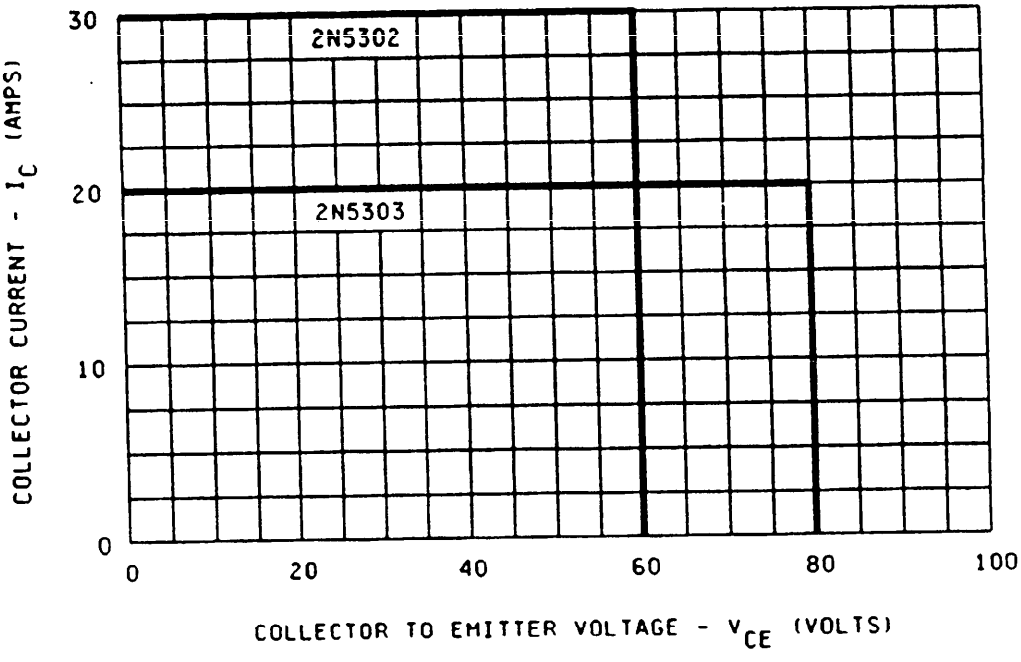


FIGURE 5. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, MC, CG,
Air Force - 13, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

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